Interfaces Bridging Embodied Artificial Intelligence and Assistive Technology

A UIST 2026 Workshop Proposal

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Abstract

Assistive robots can support people with disabilities across mobility, manipulation, communication, rehabilitation, and daily living, but their real-world impact depends on accessible, usable, and empowering interaction design. This workshop centers interface innovation for assistive human-robot interaction and asks how UIST-style techniques can improve control, personalization, safety, trust, and disabled-user agency. We propose a highly interactive full-day format with lightning talks, demos, breakout sessions, and hands-on prototyping to build a shared research agenda across UIST, accessibility, HRI, robotics, and assistive technology communities.

CCS Concepts

• **Computer systems organization → Robotics**; *Embedded and cyber-physical systems*; • **Human-centered computing → Accessibility**; • **Computing methodologies →** *Artificial intelligence.*

Keywords

intelligent user interfaces, embodied interaction, embodied AI, multimodal interaction, human–robot interaction, cyber–physical systems, assistive robotics, assistive technology, human augmentation

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1 Workshop Motivation and Scope

Assistive robots increasingly promise to support people with disabilities in activities such as mobility, manipulation, feeding, navigation, communication, rehabilitation, and daily living. Yet the success of these systems depends not only on robot autonomy, sensing, or control, but also on the interfaces through which people understand, instruct, correct, personalize, and trust them. Many assistive robots remain difficult to use because their interaction models are designed around technical capabilities rather than disabled users' lived needs, preferences, agency, and access constraints.

[5] - AccessTeleopKit: A Toolkit for Creating Accessible Web-Based Interfaces for Tele-Operating an Assistive Robot

[3] - MobiSys'22 Paper (Harish)

[2] - MobiCom'24 Paper (Harish)

[8] - Augmented Reality and Robotics: A Survey and Taxonomy for AR-enhanced Human-Robot Interaction and Robotic Interfaces

[9] - Research status of elderly-care robots and safe human-robot interaction methods

[7] - Graphene-based dual-function acoustic transducers for machine learning-assisted human–robot interfaces

[1] - Augmented reality user interface design and experimental evaluation for human-robot collaborative assembly

[6] - An analysis of design recommendations for socially assistive robot helpers for effective human-robot interactions in senior care

[4] - Touch Technology in Affective Human–, Robot–, and Virtual–Human Interactions: A Survey

This workshop asks: **How can UIST-style interaction techniques make assistive robots more usable, accessible, controllable, and empowering for people with disabilities?** We bring together researchers across user interfaces, accessibility, human-robot interaction, assistive technology, rehabilitation, robotics, AI, AR/VR, wearables, and participatory design. The workshop focuses on interface questions around assistive human-robot interaction, including speech and natural-language control, multimodal input, shared autonomy, adaptive interfaces, wheelchair-based interaction, robot teleoperation, AR/VR robot interfaces, explanation and feedback, personalization, safety, trust, and disabled-user agency.

2 Workshop Program Format

We propose a highly interactive full-day workshop for approximately 25–50 participants. The format emphasizes discussion, demonstrations, brainstorming, and prototyping rather than a sequence of long talks. Planned activities include:

- **Lightning talks** introducing key challenges in accessible human-robot interaction.
- **Demo/video sessions** showcasing assistive robot interfaces, prototypes, and interaction techniques.
- **Breakout discussions** on user agency, safety, autonomy, personalization, evaluation, and accessibility.
- **Hands-on design/prototyping** in concrete assistive robotics scenarios.
- **Synthesis** to identify open research questions, collaboration opportunities, and community-building next steps.

Participants will not be required to submit long position papers. Instead, we plan to use a short interest form asking about participant background, research interests, preferred discussion topics, optional demos, and accessibility needs.

2.1 Proposed Schedule Plan

- **9:00–9:20** Welcome, goals, and participant introductions.
- **9:20–10:10** Lightning talks on challenges in robotics, accessibility, and UI design.
- **10:30–11:30** Demo/video session on assistive robot interfaces and prototypes.
- **11:30–12:15** Breakout 1: What makes current robot interfaces inaccessible?
- **1:30–2:15** Panel: autonomy, agency, safety, and disabled-user control.
- **2:15–3:15** Hands-on prototyping for assistive robot interface scenarios.
- **3:30–4:15** Breakout 2: Future research agenda and community building.
- **4:15–5:00** Group synthesis, next steps, and collaborative outputs.

3 Intended Audience and Outcomes

The intended audience includes UIST researchers, accessibility researchers, roboticists, HRI researchers, disabled researchers and designers, assistive technology practitioners, students, and industry researchers interested in interactive systems for robotic assistance.

Expected outcomes include:

(1) A shared research agenda for accessible robot interfaces.
(2) A collection of design challenges and scenarios.
(3) New collaborations across UIST, accessibility, and robotics communities.
(4) A public workshop website collecting resources, demos, participant interests, and discussion outcomes.

4 Acknowledgments

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A Organizers

Hamza El Alaoui is a Ph.D. student in Carnegie Mellon's School of Computer Science. His research focuses on cyber-physical human augmentation, combining embedded systems, machine learning, robotics, and human–computer interaction to create systems that extend human communication, perception, and physical capabilities. His work spans speech and audio interfaces, multimodal sensing, intelligent hardware, human–AI collaboration, and human–robot integration. He previously worked on machine learning at Oracle and was a visiting scientist at Sony Computer Science Laboratories, where he worked on novel human augmentation technologies.

Harish Ram Nambiappan received his Ph.D. in Computer Science from The University of Texas at Arlington. His research focuses on developing assistive navigation and human robot interactive systems that enable blind and low-vision people to navigate dynamic indoor environments and interact directly with robots. He has built cross-modal interfaces through which users can locate objects, direct robotic manipulation, and remain safe around moving robots and people. His broader work also includes autonomous service robots for healthcare, placing his research at the intersection of assistive technology, robot autonomy, and human–robot interaction.

Atieh Taheri is a Presidential Postdoctoral Fellow in Carnegie Mellon's HCI Institute. Her research combines accessibility, human-centered AI, mixed reality, and participatory design to build interactive systems that expand agency and creative expression for people with disabilities. Her work includes hands-free controllers, tactile interfaces, accessible conversational systems, VR experiences, and accessible generative-AI tools. She previously conducted computer vision, machine-learning, XR, and accessibility research at Apple, Magic Leap, and Google. She earned her Ph.D. in Electrical & Computer Engineering from the University of California, Santa Barbara.

Max Pascher is a Postdoctoral Researcher at TU Dortmund University and a guest researcher at the University of Duisburg–Essen. He received his Ph.D. in Human–Robot Interaction from the University of Duisburg–Essen. His research focuses on AI-enhanced human–robot collaboration, particularly shared-control techniques for assistive robotic arms, intervention strategies for autonomous robot tasks, and XR-based tools for prototyping and evaluating human–robot interfaces. He also works on multimodal interaction, distributed systems, and methods for keeping users meaningfully in control of increasingly autonomous robotic systems.

Vinitha Ranganeni is the Human–Robot Interaction Lead at Hello Robot, Inc, where she develops and evaluates human-centered capabilities for the Stretch mobile manipulator. She received her Ph.D. in Computer Science and Engineering from the University of Washington, where she created AccessTeleopKit, an open-source toolkit for building customizable teleoperation interfaces for assistive robots. Her work also includes long-term in-home deployments with people with motor impairments, accessible robot control, shared autonomy, navigation, and manipulation.

Stephanie Valencia is an Assistant Professor in the College of Information at the University of Maryland. Her research focuses on equitable access to assistive technology, open-source hardware, and the design of communication systems that increase user agency, accessibility, and enjoyment. Using participatory design, she develops AI-enabled and embodied expressive technologies for augmentative and alternative communication users, with an emphasis on building, deploying, and rigorously evaluating systems in real-world settings. She received her Ph.D. from Carnegie Mellon University's Human–Computer Interaction Institute.

Tapomayukh "Tapo" Bhattacharjee is an Assistant Professor of Computer Science at Cornell University, where he directs the EmPRISE Lab. He received his Ph.D. in Robotics from the Georgia Institute of Technology and was an NIH Ruth L. Kirschstein NRSA Postdoctoral Research Associate at the University of Washington. His research focuses on physical robot caregiving and physical human–robot interaction, particularly assistive robots that support people with mobility limitations in activities of daily living. His lab takes a full-stack robotics approach spanning sensing, perception, planning, learning, control, manipulation, haptic perception, and tactile sensing, with an emphasis on building and evaluating complete robotic systems in real-world settings.

Eshed Ohn-Bar is an Assistant Professor of Electrical & Computer Engineering and affiliated faculty in Computer Science at Boston University, where he leads the Human-to-Everything Lab. He received his Ph.D. in Electrical Engineering from the University of California, San Diego, and was a Humboldt Fellow at the Max Planck Institute. His research develops machine-intelligence systems for real-world embodied, assistive, and autonomous applications, spanning perception, learning, planning, control, and human–machine interaction. His work includes autonomous-driving systems, accessible navigation technologies, and AI agents that provide adaptive, real-time assistance to blind and low-vision people.

Jeffrey P. Bigham is an Associate Professor in Carnegie Mellon's School of Computer Science in the Human-Computer Interaction Institute and Language Technologies Institute. He also serves as the Director of Human-Centered Machine Intelligence and Responsible AI at Apple Inc, where he leads research on AI/ML design, system behavior, accessibility technologies, learning sciences, and computational understanding of user interfaces. His research combines machine learning, speech and language technologies, computer vision, interactive systems, and real-time human computation to build technically robust and deployable systems, particularly for accessibility and assistive technology. He received his Ph.D. in Computer Science & Engineering from the University of Washington.